

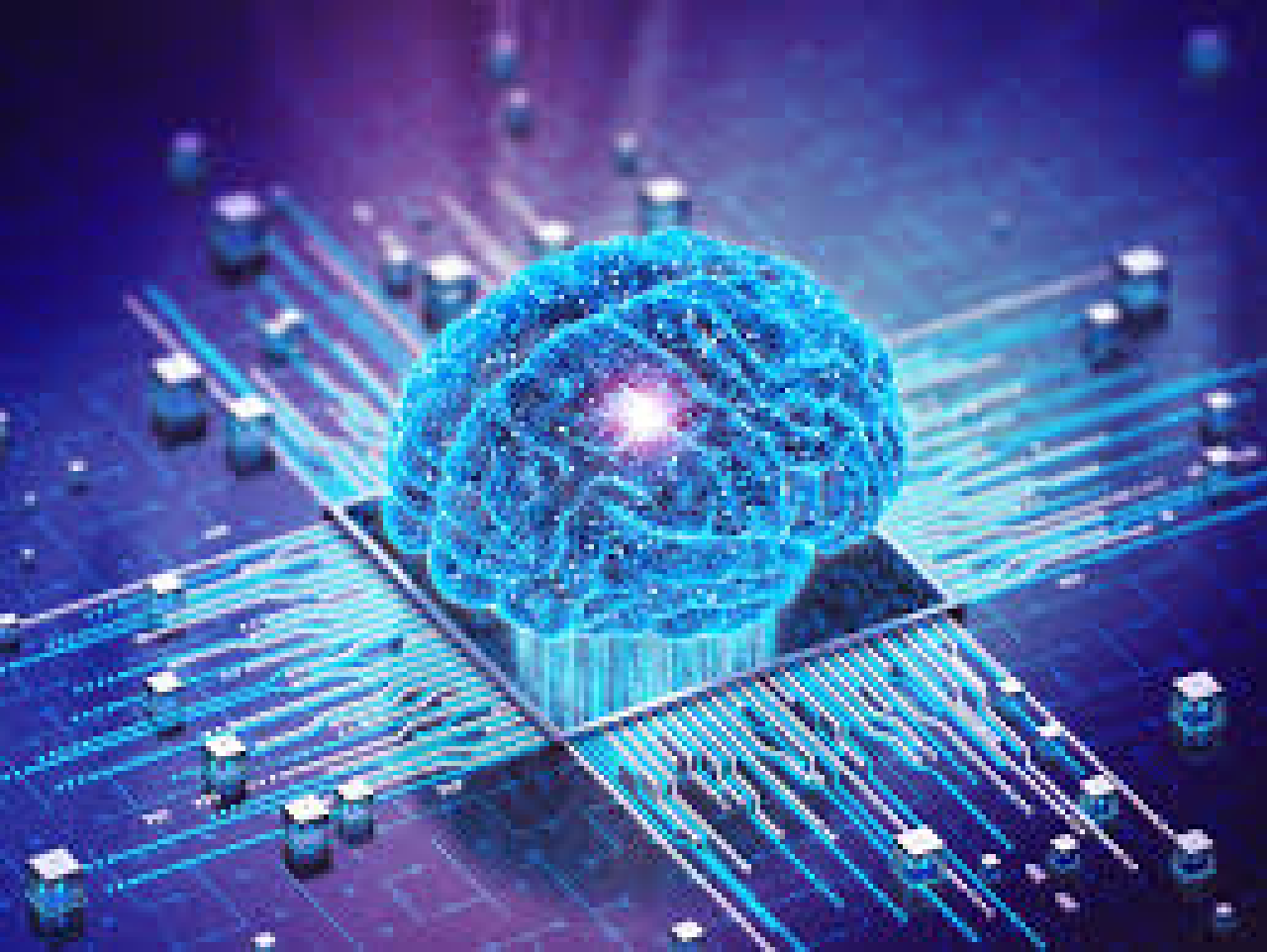
Edu Fact – AI Hallucination Corrector for Educational Content

Haneesh Ponam

Harshitha Tankasala

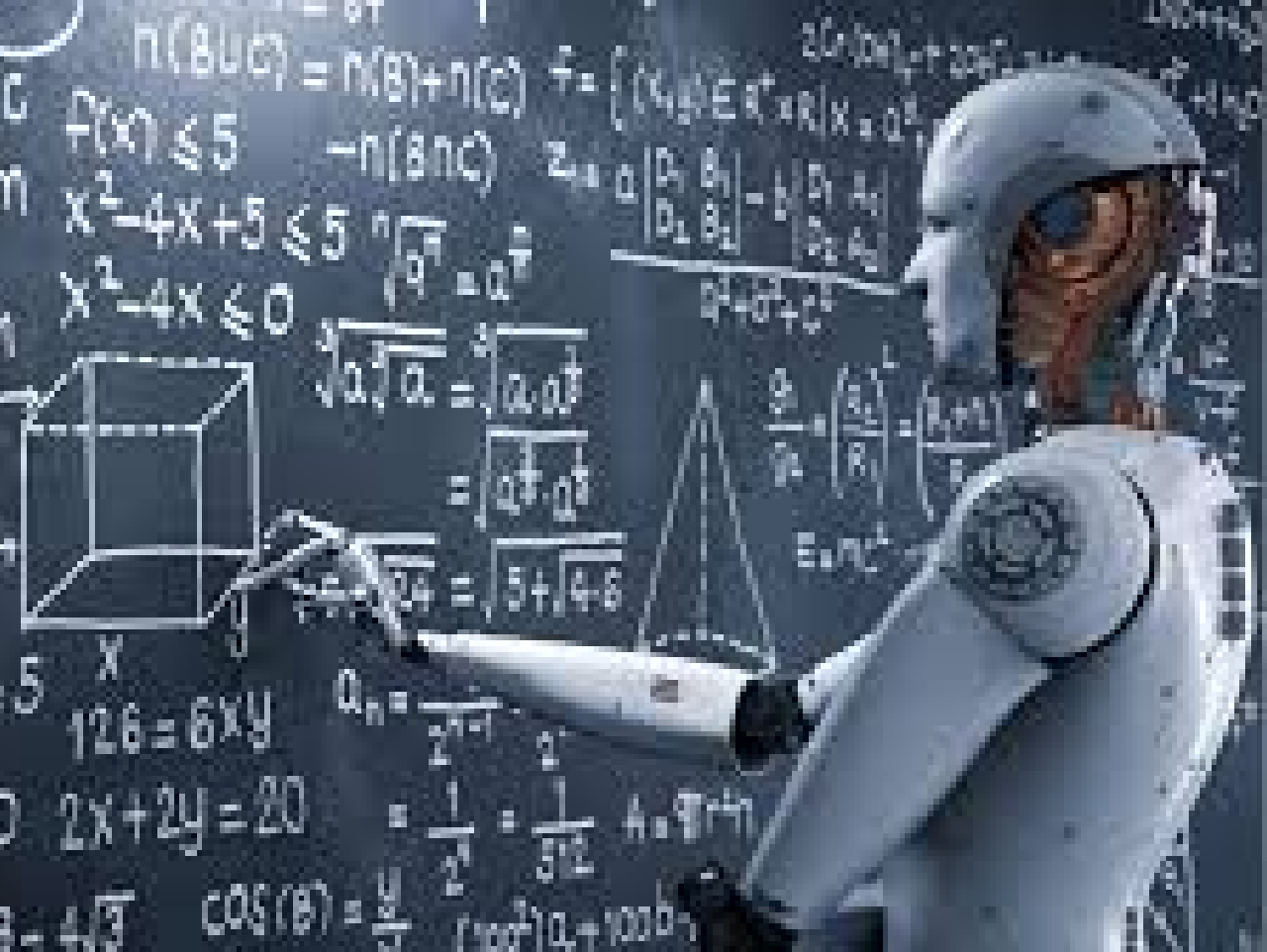
Andrea Domo-Ibea





- Students rely on AI and unknowingly absorb misinformation
- 30–50% of generative AI answers contain factual errors
- Multilingual misinformation is unaddressed
- Teachers can't manually verify every student's AI-assisted work

The Problem: AI Hallucinations in Education



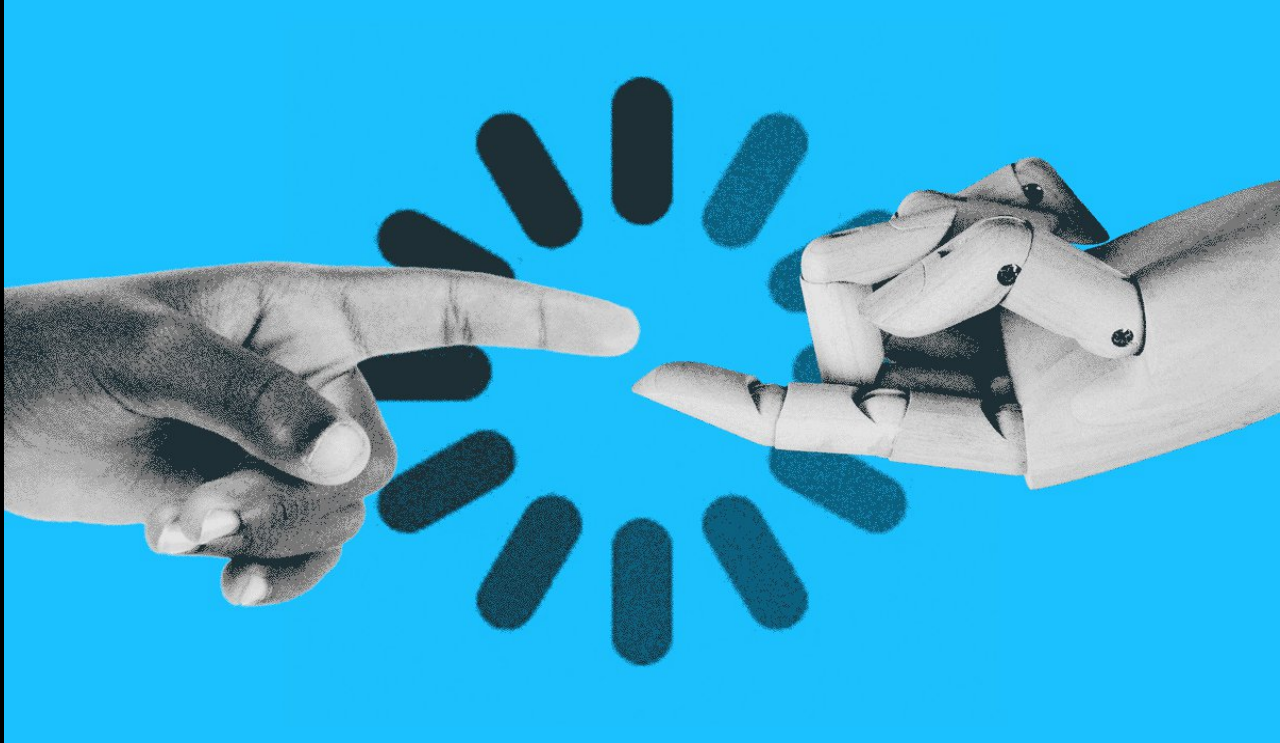
- **67% of U.S. students** use AI tools weekly for schoolwork (Pew Research, 2024).
- **1 in 3 students** say AI has given them false information that harmed their grade.
- Misinformation spreads faster in multilingual classrooms due to translation hallucinations.
- Schools lack standardized AI-literacy verification tools.

Why It Matters

Current State Summary

- Students use Google, GPT, and Wikipedia with **no verification layer**.
- AI fact-checking tools do not extract claims.
- No real-time, multilingual truth scoring exists for education.
- Teachers rely on manual review → slow, inconsistent, prone to human error.





Porter's Five Forces — Current Industry

- **Buyer Power: HIGH**
- Students/teachers have unlimited free tools; switching cost = 0.
- **Threat of Substitutes: VERY HIGH**
- Manual fact-checking, ChatGPT, Grammarly, Google Search.
- **Threat of New Entrants: HIGH**
- Open-source LLMs make it easy to build simple tools.
- **Supplier Power: MEDIUM**
- Accuracy is dependent on knowledge bases (Wikidata, Google Fact Check APIs).
- **Industry Rivalry: MEDIUM-HIGH**
- Many EdTech tools compete for classroom attention.

Current Value Stream

- **Student → AI output → “confident but wrong” answers → no verification → misinformation used in assignments → teachers catch errors too late**



Customer Journey (Before EduFact)

01

Student asks AI
→ receives
answer →
assumes
correctness

02

No claim
breakdown

03

No multilingual
accuracy

04

Teacher receives
work → must
manually validate
information

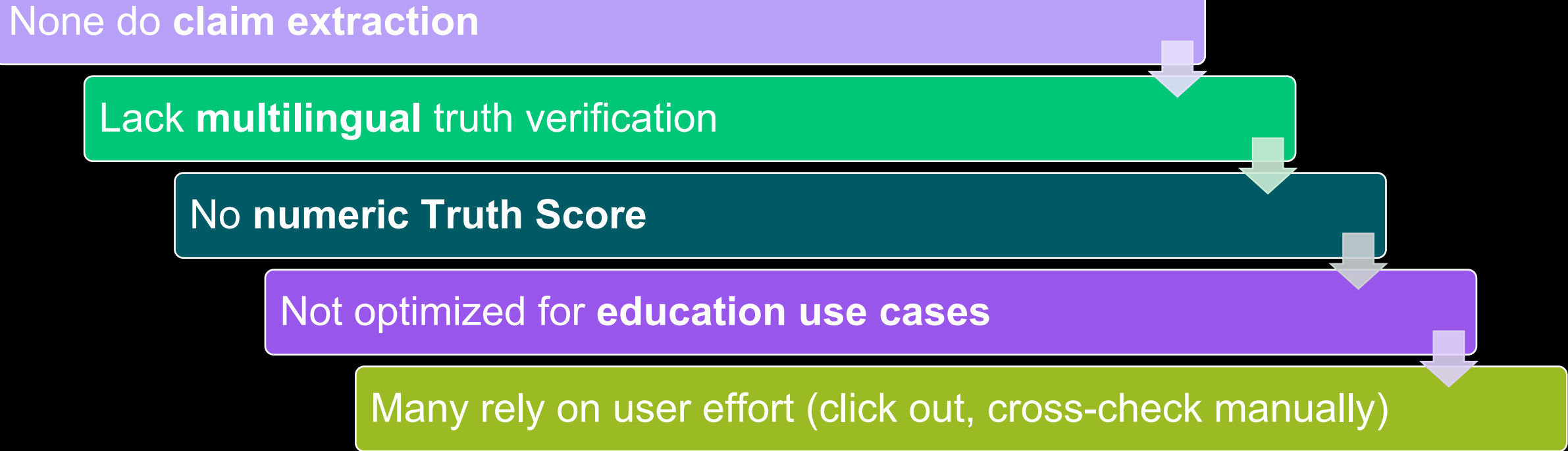
Options Considered

- Google Fact Check Explorer
- Manual research through sources
- Browser plugins (FactCheck.org, extensions)
- ChatGPT "with citations"
- GrammarlyGO / Bing Chat (limited fact-check mode)



Why Current Solutions Fail

None do **claim extraction**



```
graph TD; A[None do claim extraction] --> B[Lack multilingual truth verification]; B --> C[No numeric Truth Score]; C --> D[Not optimized for education use cases]; D --> E[Many rely on user effort (click out, cross-check manually)]
```

Lack **multilingual** truth verification

No **numeric Truth Score**

Not optimized for **education use cases**

Many rely on user effort (click out, cross-check manually)

EduFact Overview

- A multilingual AI truth-scoring tool that extracts claims, verifies them, and outputs transparent “Truth Scores” for each claim.

What Makes EduFact a Digital Innovation

- Radical shift from “AI confidence” → AI truth transparency
- Real-time claim extraction
- Multilingual verification (Japanese, Filipino, Telugu tested)
- Designed specifically for students, teachers, and academic integrity
- Demo Results (Real EduFact Tests)
- (Japanese Article Test) → all Truth Scores = 0
(Filipino Article Test) → all Truth Scores = 0
(Telugu Article Test) → all Truth Scores = 0
→ Demonstrates cross-lingual factual ground



Key Features

- Automated claim extraction
- Multilingual truth evaluation
- Numeric truth score (0–100)
- Teacher dashboard (future)
- Simple student-friendly UI

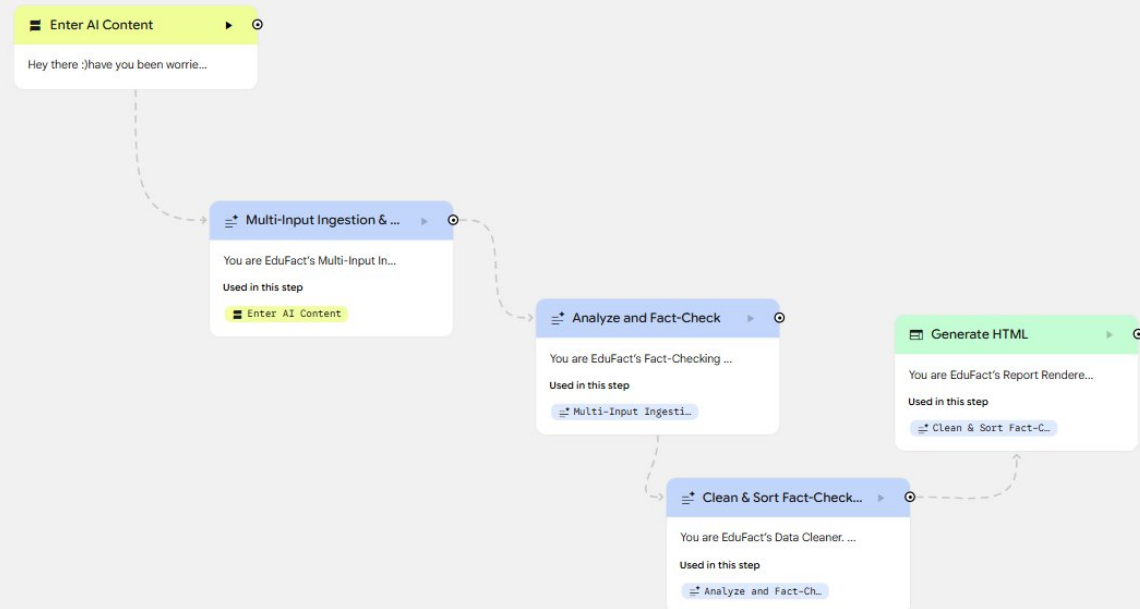


System Architecture

- UI Layer → NLP Claim Extraction → Language Detection → Normalization → Knowledge Graph Verification → Truth Score Engine → Visualization Layer



EduFact – AI Fact-Checking Pipeline Architecture



Frontend (App UI)

- User pastes AI-generated content into the “**Enter AI Content**” block.

1 Multi-Input Ingestion & Normalization

- Cleans the raw text (removes noise, duplicates, formatting junk).
- Merges everything into one coherent **Study Text**.

2 Analyze & Fact-Check

- LLM reads the full Study Text.
- Extracts the most important factual claims.
- For each claim, assigns a **verdict** and **Truth Score (0–100)** and suggests sources.

3 Clean & Sort Fact-Checks

- Validates the JSON structure.
- Removes duplicates / non-factual lines.
- Sorts claims from **most risky (low score)** to **most reliable (high score)**.

4 Generate HTML Report

- Turns the cleaned JSON into a **visual dashboard**.
- Shows traffic-light colors (, summary counts, and a card for each claim with sources.

Result:

- A shareable, human-friendly report that shows **how accurate the AI content is** and **which claims need attention**.

How to Use EduFact

1

Open the Website.

2

Login using
Google.

3

Paste Your AI
generated content.

4

Click Check
content.

5

Wait until it checks
the content and
gives you the truth
score.

Future State: Porter's Five Forces (After EduFact Adoption)

1. Threat of New Entrants — LOW

- Requires advanced LLM evaluation, RAG pipelines, and high-quality data
- Strong first-mover advantage in “AI truth verification” category
- High technical complexity reduces easy replication

2. Supplier Power — MEDIUM

- Today: dependence on major AI providers (OpenAI, Google, Anthropic)
- Future: multi-model backend reduces dependency (swap models easily)
- Ability to negotiate volume pricing as user base grows

3. Buyer Power — LOW

- Once integrated into LMS (Canvas/Blackboard/Google Classroom), switching costs increase
- Teachers/universities depend on stable verification reports
- Creates institutional stickiness and reduces churn

4. Threat of Substitutes — LOW

- No current tool automates **claim extraction + truth scoring + source verification**
- Google Search and Perplexity cannot produce structured “Truth Reports”
- Human verification is too slow and inconsistent to be a real alternative.

5. Industry Rivalry — LOW

- EduFact establishes a **new category**: AI Accuracy-as-a-Service
- Plagiarism/AI detection tools (Turnitin, GPTZero) target different problems
- Minimal direct competition in factual verification for education

Future-State Value Stream (with EduFact):

- **After adopting EduFact, the verification workflow becomes instant, linear, and frictionless — eliminating delays, manual searching, and inconsistent fact-checking effort.**

New Process Steps (Future State)

- AI Content is Generated
- User pastes content into EduFact
- One-click Fact Check
- Automated Claim Extraction
- Truth Scoring + Source Verification
- Instant HTML Truth Report
- User reviews accurate, color-coded insights.

Benefits of the New Flow

- Verification time drops from 10–20 minutes → 3 seconds
- Removes dependency on manual Google searching
- Uniform fact-checking improves academic reliability
- Students & educators trust AI content more
- Creates a clean, standardized accuracy report

Business Model Strategy



Freemium Model:

- Free tier: 20 checks/month
- Pro tier: unlimited checks, PDF reports, bulk uploads
- B2B SaaS: APIs for ed-tech platforms (Coursera, Chegg, Notion, Perplexity)
- Chrome extension for student and educator market

Revenue Streams:

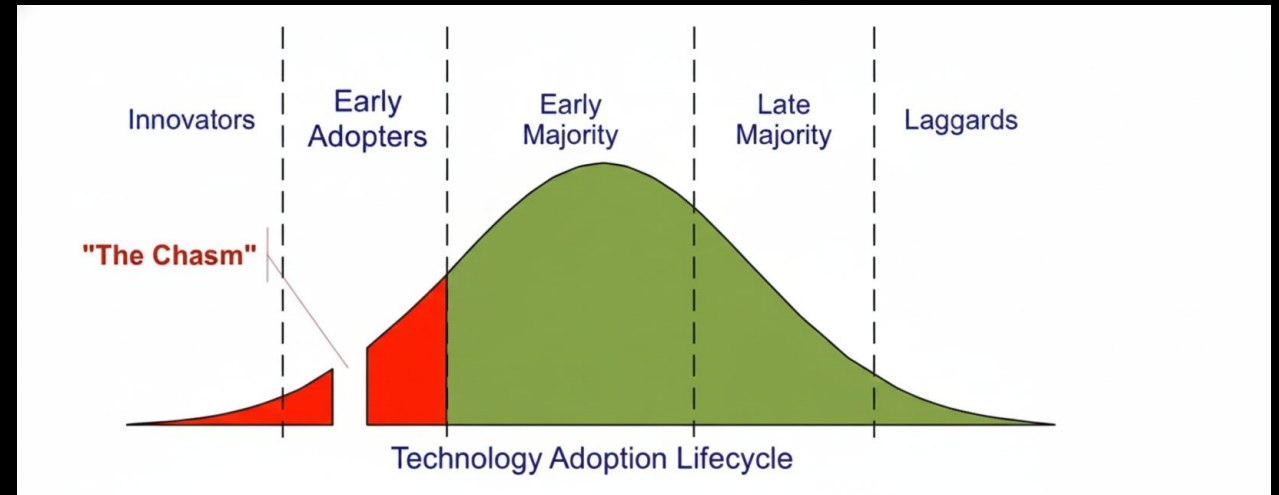
- Subscriptions
- University licenses
- API metered usage
- LMS integration fees

Network Effects

- **1. Data Network Effect:**
More users → more fact-checks → better detection → higher accuracy → attracts more users
- **2. Platform Network Effect:**
Students → teachers → universities → EdTech platforms → publishers
- **3. Integration Network Effect:**
More LMS / extension integrations → higher switching cost → market dominance

Technology Adoption Lifecycle

- **Innovators:** AI enthusiasts, tech students
- **Early Adopters:** STEM majors, researchers
- **Early Majority:** universities, tutoring platforms
- **Crossing the Chasm Strategy:** integrate EduFact into LMS + Chrome extension for easy one-click adoption
- **Late Majority:** general student population
- **Laggards:** schools resisting AI



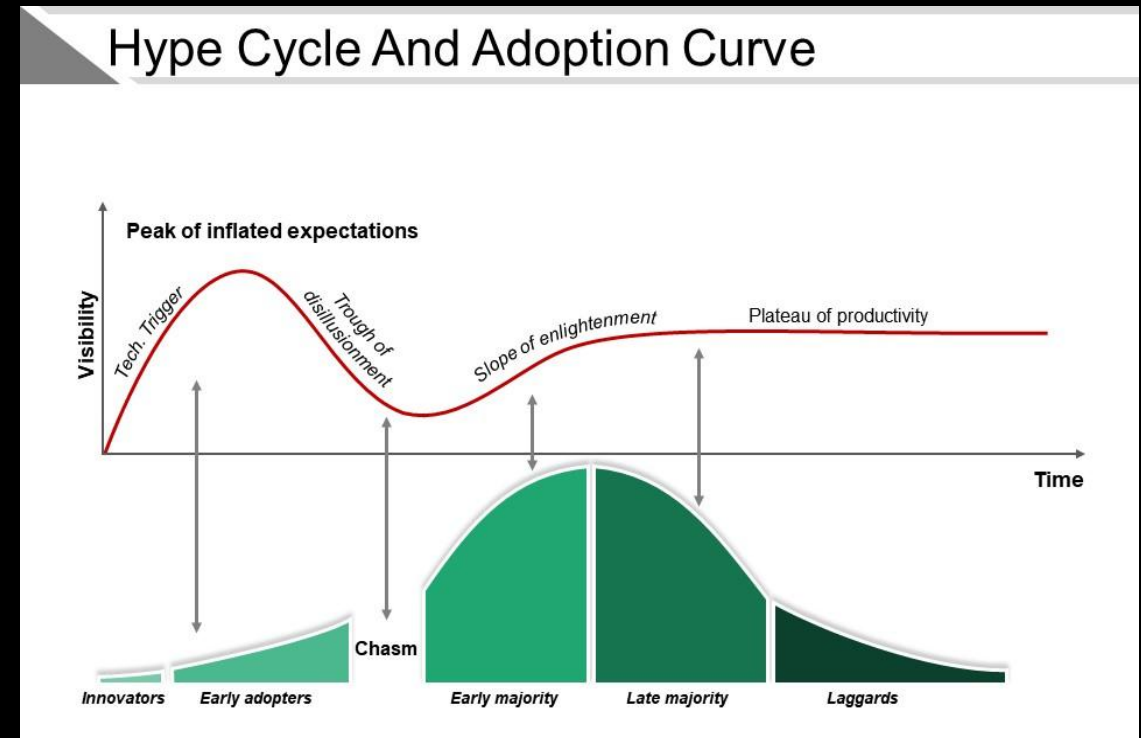
EduFact on the AI Verification Hype Cycle (Emerging Technologies 2025)

EduFact is currently between:

“Technology Trigger → Peak of Inflated Expectations”

Meaning:

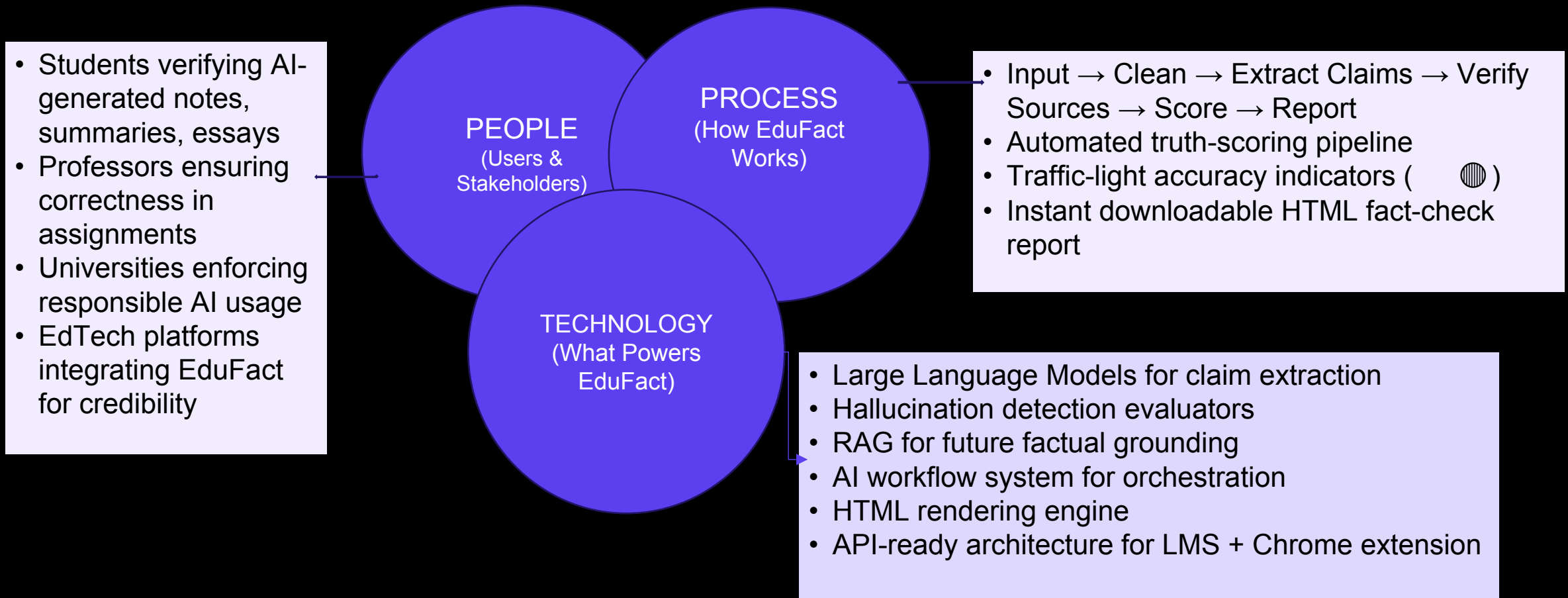
- The problem (AI hallucination) is HUGE
- The market has NO solutions
- EduFact is early → perfect for innovation funding
- High visibility + low competition = Blue Ocean



EduFact Scaling Plan



PEOPLE, PROCESS, TECHNOLOGY FRAMEWORK



User Testing Insights & Prototype Evaluation (Real Feedback Analysis)

1. Overall User Experience

- Users found the app **simple, intuitive, and easy to test.**
- The **traffic-light truth meter** was the most liked feature (immediately understandable).
- Most users said EduFact feels **useful for studying, assignments, and validating AI answers.**

2. Key Strengths Identified

- **Fast + automatic fact-checking**
- **Clear scoring system (0–100)**
- **Clean HTML report** with structured claims and sources
- **Novel concept** — no other tool checks factual accuracy of AI content

User Testing Insights & Prototype Evaluation (RealFeedback Analysis)-Continued

3 .Pain Points / Issues Noticed

- Response speed was slow for longer text inputs
- Some reports were confusing for non-technical users
- Download button did not work consistently
- Limited to text-only input (no PDFs, images, or voice uploads yet)

4. What Users Want Next (Feature Req

- 4. What Users Want Next (Feature Req
- A Chrome extension to verify content directly on websites
- Ability to share reports with classmates or teachers
- A “Quick Truth Score Only” mode for instant checks

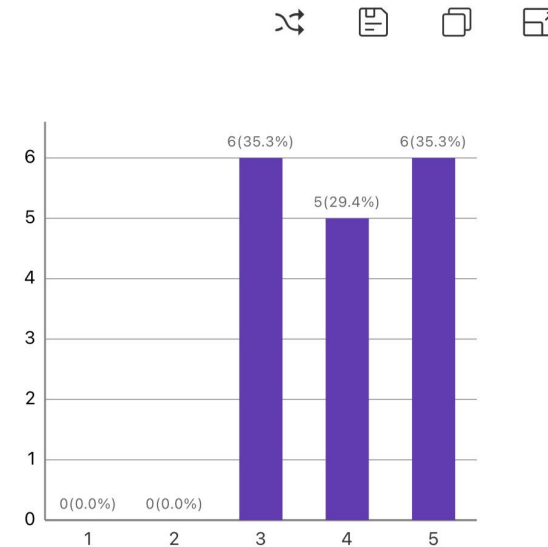
5. Conclusion & Next Steps

- User feedback confirms strong demand for an AI truth-verification tool.
- The prototype is promising, but needs improvements in speed, UX clarity, and input options before scaling.

Prototype User Results Analysis(MARKET RESEARCH):

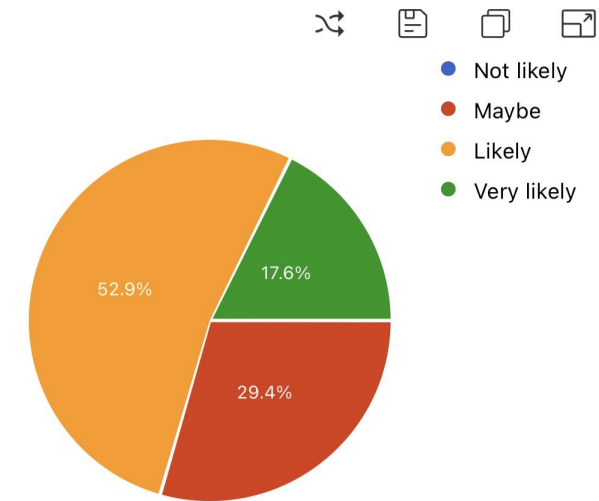
How easy was it to use EduFact?

17 Responses



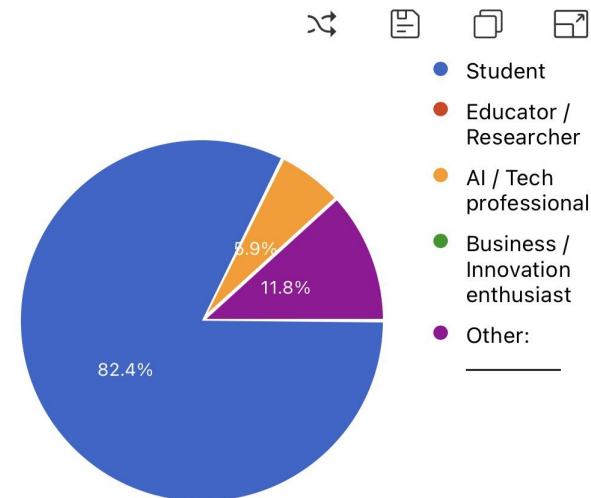
How likely are you to use EduFact again for coursework or teaching?

17 Responses



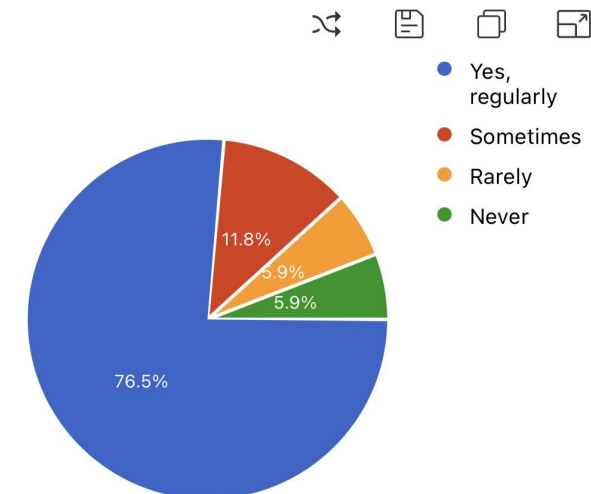
Your role / background

17 Responses

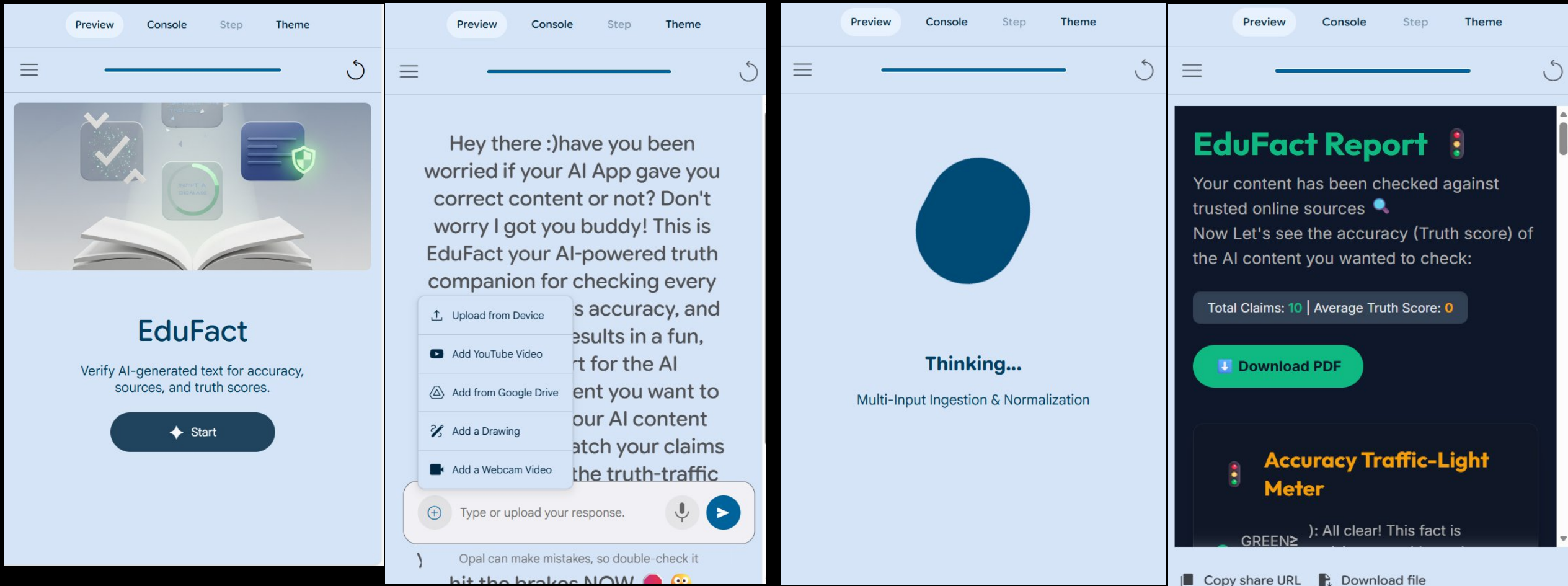


Have you used AI tools for learning/writing (ChatGPT, Gemini, Perplexity, Notion AI)?

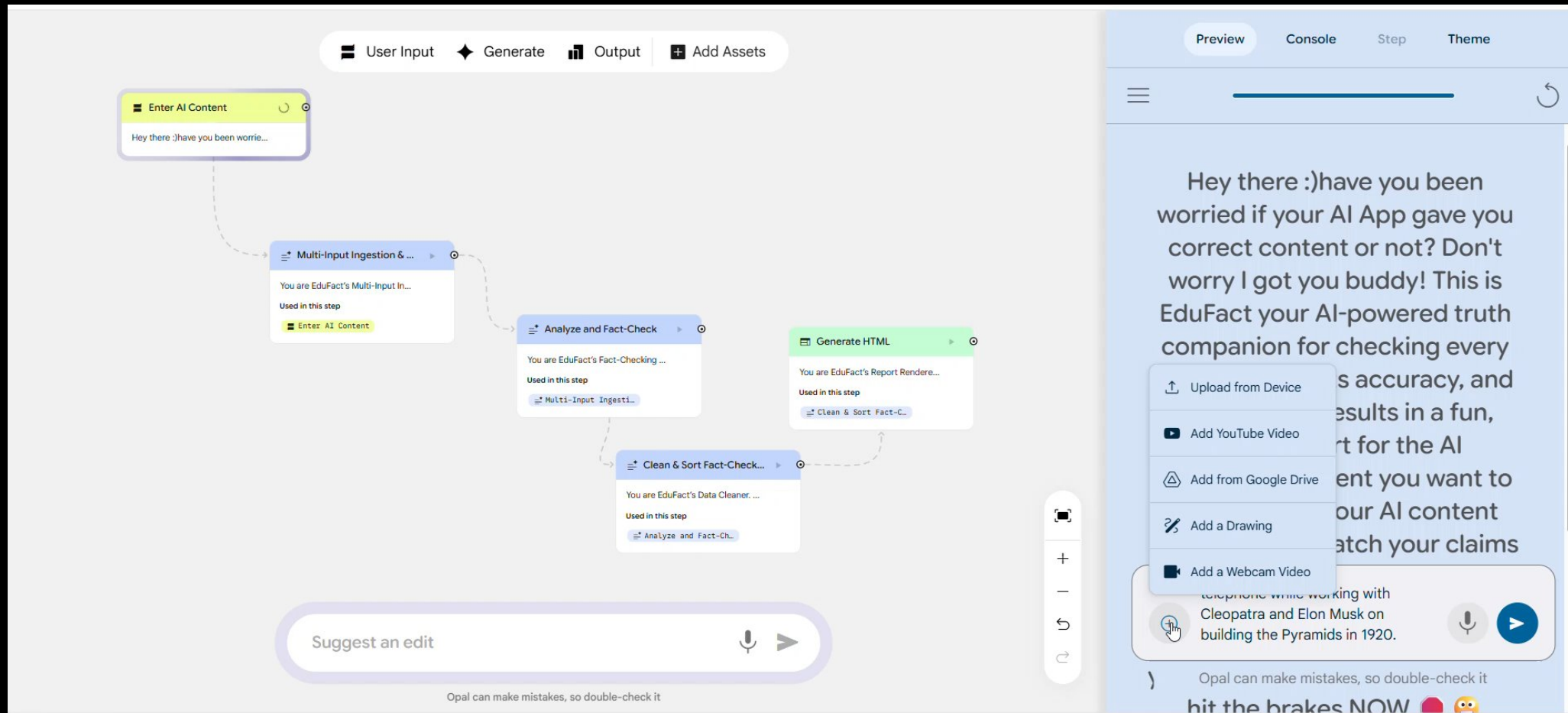
17 Responses



EDUFACT FRONTEND DESIGN



EDUFACT DEMO/LIVE WORKFLOW



CONCLUSION-*Why EduFact Should Be Funded*

- Solves a real, growing academic problem
- First-mover advantage in AI truth verification
- Strong business model with SaaS + academic partnerships
- Scalable architecture
- Clear market demand + network effects
- Ready prototype demonstrates feasibility
- **Tagline:**
- *“EduFact — Because even AI needs a fact-check.”*

REFERENCES

- ARTICLES:
- 1. OpenAI – “GPT Models and Hallucinations”
 - <https://platform.openai.com/docs/guides/hallucinations>
- 2. Google DeepMind – “Reducing AI Hallucinations with Reinforcement Learning and Verification”
 - <https://deepmind.google/discover/blog>
- 3. Vectara – “Hallucination Evaluation Model (HHEM)”
 - <https://vectara.com/hallucination-leaderboard/>
- 4. Anthropic – “Hallucinations in Large Language Models”
 - <https://www.anthropic.com/index/research>
- 5. Stanford HAI – “Evaluating Accuracy & Hallucinations in LLMs”
 - <https://hai.stanford.edu/research>
- 6.MARKET RESEARCH WITH REAL USERS COLLECTED IN GOOOGLE FORMS
- 7.CLASS MATERIALS
- 8.GOOGLE'S LLM GUIDE BOOK
- APP LINK: [TRY IT HERE](#)

THANK YOU

"If this presentation had any hallucinations... don't worry, EduFact will catch them."